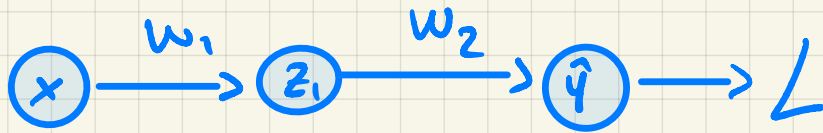



BACKPROPAGATION & THE CHAIN RULE



- GOAL: Cambiare w_2 per min L .
- P.MA: L non dipende direttamente da w_2
- SOLUZ: L **CHAIN RULE** (derivata di f. di f.)

CHAIN RULE:

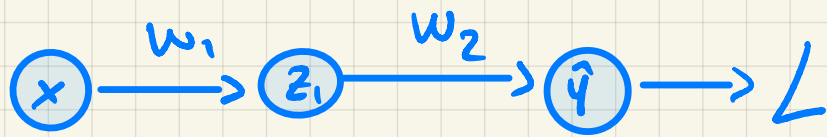
$$\Rightarrow \frac{\partial g(\overbrace{f(x)}^y)}{\partial x} = \frac{\partial g}{\partial y} \cdot \frac{\partial f(x)}{\partial x}$$

EX:

$$g(x) = \log(\overbrace{x^3}^y) + 20$$

$$\frac{\partial g(x)}{\partial x} = \frac{\partial [\log(y) + 20]}{\partial y} \cdot \frac{\partial x^3}{\partial x} = \frac{1}{y} \cdot (3 \cdot x^2) =$$
$$= \frac{1}{x^3} \cdot 3x^2 = \frac{3}{x}$$

$$\frac{\partial L(w)}{\partial w_2} = \frac{\partial L(w)}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2}$$



- GOAL: cambiare w_1 per min L .

$\Rightarrow w_1$ ancora più aumentato
 ma con la chain rule
 arriviamo ad esplicitare la
 dipendenza:

$$\frac{\partial L(w)}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

GRADIENT DESCENT OPTIMIZATION ALGORITHMS

RECALL

SGD:

$$\Theta_t = \Theta_{t-1} - \gamma \frac{\partial L(\Theta)}{\partial \Theta}$$

• MOMENTUM

→ SMORZA (dampen) le oscillazioni,

RALENTANDO quando il gradiente cambia direz.

→ Accelera SGD se la direzione si mantiene nel tempo

$$1) \quad v_t = \gamma \frac{\partial L_t(\Theta)}{\partial \Theta} + \gamma v_{t-1}$$

$$2) \quad \Theta_t = \Theta_{t-1} - v_t =$$

$$= \Theta_{t-1} - \left[\gamma \frac{\partial L}{\partial \Theta} + \gamma v_{t-1} \right]$$

$$\left[\gamma \frac{\partial L}{\partial \Theta} + \gamma v_{t-2} \right]$$

...

$$\Theta_t = \Theta_{t-1} - \nu_t =$$

$$\boxed{\nu_t = \gamma \frac{\partial L_t(\theta)}{\partial \theta} + \gamma \nu_{t-1}}$$

$$= \Theta_{t-1} - \left[\gamma \frac{\partial L_t(\theta)}{\partial \theta} + \gamma \nu_{t-1} \right] =$$

$$= \Theta_{t-1} - \gamma \frac{\partial L_t(\theta)}{\partial \theta} - \gamma \left[\gamma \frac{\partial L_{t-1}(\theta)}{\partial \theta} + \gamma \nu_{t-2} \right] =$$

$$= \Theta_{t-1} - \gamma \frac{\partial L_t(\theta)}{\partial \theta} - \gamma^2 \frac{\partial L_{t-1}(\theta)}{\partial \theta} - \gamma^2 \nu_{t-2} =$$

$$= \Theta_{t-1} - \gamma \frac{\partial L_t(\theta)}{\partial \theta} - \gamma^2 \frac{\partial L_{t-1}(\theta)}{\partial \theta} - \gamma^2 \left[\gamma \frac{\partial L_{t-2}(\theta)}{\partial \theta} + \gamma \nu_{t-3} \right] =$$

$$= \Theta_{t-1} - \underbrace{\gamma \frac{\partial L_t(\theta)}{\partial \theta}}_{(*)} - \underbrace{\gamma^2 \frac{\partial L_{t-1}(\theta)}{\partial \theta}}_{(*)} - \underbrace{\gamma^3 \frac{\partial L_{t-2}(\theta)}{\partial \theta}}_{(*)} - \gamma^3 \nu_{t-3}$$



- 1) Se le derivate (*) sono coerenti si ha ACCELERAZ.
- 2) γ ha esponente via via crescente, quindi i ν diventano via via meno influenti.

$$\boxed{\Theta_t = \Theta_{t-1} - \gamma \cdot \sum_{i=0}^{\infty} \gamma^i \cdot \frac{\partial L_{t-i}(\theta)}{\partial \theta}}$$

• ADAM.

1st MOMENT ESTIMATE $\rightarrow m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$

2nd MOMENT ESTIMATE $\rightarrow v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$

where: $g_t = \frac{\partial L}{\partial \theta_t}$

PROBLEM: m_t and v_t biased toward 0 especially in the first steps

UNBIASED 1st MOM. EST $\rightarrow \hat{m}_t = \frac{m_t}{1 - \beta_1^t}$

UNBIASED 2nd M.E. $\rightarrow \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$

Bias Corrected 1st and 2nd MOMENTS.

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t$$

INIT:

$\beta_1 = 0.9$; $\beta_2 = 0.999$; $\epsilon = 10^{-8}$
 $m_0 = 0$; $v_0 = 0$